

Enhancing Rule Construction in Grammatical Evolution with Advanced Crossover Techniques

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Abstract

Many real-world problems can be formulated as classification tasks in machine learning and addressed using a wide range of advanced algorithms developed for this purpose. Among these approaches, methods based on Grammatical Evolution (GE) can also be employed, such as GenClass, a technique designed to automatically construct classification rules. Although GenClass has been successfully applied to a variety of classification problems, its evolutionary search process may suffer from limitations such as entrapment in local minima of the error function and premature convergence, which can prevent the method from discovering high-quality classification rules. The present work proposes an enhanced version of GenClass through the introduction of advanced crossover techniques operating on the integer-valued chromosomes employed by the method, with the aim of achieving a significant reduction in classification error. The proposed crossover operators follow two different strategies. Some of them are purely stochastic, introducing greater diversity into the evolutionary search, whereas others exploit information already generated during the Grammatical Evolution process to perform targeted exchanges of genetic material between chromosomes. In this way, the latter operators attempt to guide the evolutionary search toward more promising regions of the search space while preserving useful information discovered during previous generations. The proposed techniques were evaluated on a diverse collection of real-world classification datasets, including several problems characterized by a large number of patterns and/or a large number of classes. The experimental results demonstrate that the proposed crossover strategies can substantially improve the performance of the original method, leading to a noticeable reduction in classification error across a wide range of classification problems.

Keywords: Grammatical Evolution; Classification; Crossover Operators; Evolutionary Algorithms; Machine Learning.

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1. Introduction

2. Materials and Methods

3. Results

3.1. Experimental datasets

The datasets used in the conducted experiments were downloaded from the following following online databases:

1. The UCI website, <https://archive.ics.uci.edu/> (accessed on 24 June 2026)[1]

2. The Keel website, <https://sci2s.ugr.es/keel/datasets.php> (accessed on 24 June 2026) [2].

The used datasets are the following:

1. **Alcohol**. This dataset is related to experiments on alcohol consumption [3].
2. **Appendictis** a dataset originated in [4].
3. **Beed**.
4. **Dermatology** dataset [5], a dataset that contains measurements for dermatological deceases.
5. **Gene**.
6. **Hayes-roth** dataset [6], a dataset with 3 classes. An artificial data set created to test the behaviour of prototype-based classifiers. The hobby attribute was generated at random, thus it is employed to add noise to the data.
7. **Hepatitis** dataset. This data set contains a mixture of integer and real valued attributes, with information about patients affected by the Hepatitis disease.
8. **Housevotes**, that uses data from the Congressional voting in USA [7].
9. **Ionosphere**, that used data from various measurements in the ionosphere [8,9].
10. **Lymography** [10], which is a dataset with four distinct classes.
11. **Nursery** dataset [11]. This database was derived from a hierarchical decision model originally developed to rank applications for nursery schools. It was used during several years in 1980\'s when there was excessive enrollment to these schools in Ljubljana, Slovenia, and the rejected applications frequently needed an objective explanation. The task is to rank applications for a given nursery school.
12. The **Optdigits** dataset [12], used for the recognition of handwritten digits from optical scans.
13. **Penbased** dataset. A digit data base made by collecting 250 samples from 44 writers, using only (x, y) coordinate information represented as constant length feature vectors, which were resampled to 8 points per digit (therefore the data set contains 8 points x 2 coordinates = 16 attributes). The class label represents the code of the digit written.
14. **Phoneme**, where the purpose of this dataset is to distinguish between nasal and oral sounds.
15. **PirVision** dataset [13], that contains occupancy detection data.
16. **Satimage** dataset, related to satellite images.
17. **Segment** dataset [14], which is used for image processing .
18. **Shuttle** dataset. This data set was generated originally to extract comprehensible rules for determining the conditions under which an autolanding would be preferable to manual control of a spacecraft.
19. **Spambase**, a dataset used for the detection of spam emails.
20. **Spiral**, which is an artificial dataset.
21. **Student**, a dataset related to some experiments in schools [15].
22. **Splice** dataset, which is inspired by biology.
23. **Texture** dataset. The aim in this data set is to distinguish between 11 different textures (Grass lawn, Pressed calf leather, Handmade paper, Raffia looped to a high pile, Cotton canvas, ...), each pattern (pixel) being characterised by 40 attributes built by the estimation of fourth order modified moments in four orientations: 0, 45, 90 and 135 degrees.
24. **Thyroid** dataset. This data set is one of the several databases about Thyroid available at the UCI repository. The task is to detect if a given patient is normal (1) or suffers from hyperthyroidism (2) or hypothyroidism (3).
25. **Twonorm** dataset. This is a 20 dimensional, 2 class classification problem. Each class is drawn from a multivariate normal distribution.

26. **Vehicle** dataset. The purpose is to classify a given silhouette as one of four types of vehicle, using a set of features extracted from the silhouette. The vehicle may be viewed from one of many different angles.
27. **Yeast** dataset. This database contains information about a set of Yeast cells. The task is to determine the localization site of each cell among 10 possible alternatives.
28. **EEG** dataset, which is a medical dataset related to EEG measurements[16,17]. From the original dataset the following cases were obtained: Z_F_S, ZO_NF_S, Z_O_N_F_S and ZONF_S.

Table 1. The main characteristics of each dataset.

DATASET	PATTERNS	CLASSES
ALCOHOL	476	4
APPENDICITIS	106	2
BEED	8000	4
DERMATOLOGY	358	6
GENE	3175	3
HAYES ROTH	160	3
HEPATITIS	80	2
HOUSEVOTES	232	2
IONOSPHERE	351	2
LYMOGRAPHY	148	4
NURSERY	12690	5
OPTDIGITS	5620	10
PENBASED	10992	10
PHONEME	5404	2
PIRVISION	15302	3
SATIMAGE	6435	7
SEGMENT	2310	7
SHUTTLE	58000	7
SPAMBASE	4597	2
SPIRAL	2000	2
SPLICE	3190	3
STUDENT	403	4
TEXTURE	5500	11
THYROID	7200	3
TWONORM	7400	2
VEHICLE	846	4
YEAST	1484	10
Z_F_S	300	3
Z_O_N_F_S	500	5
ZO_NF_S	500	3
ZONF_S	500	2

3.2. Experimental results

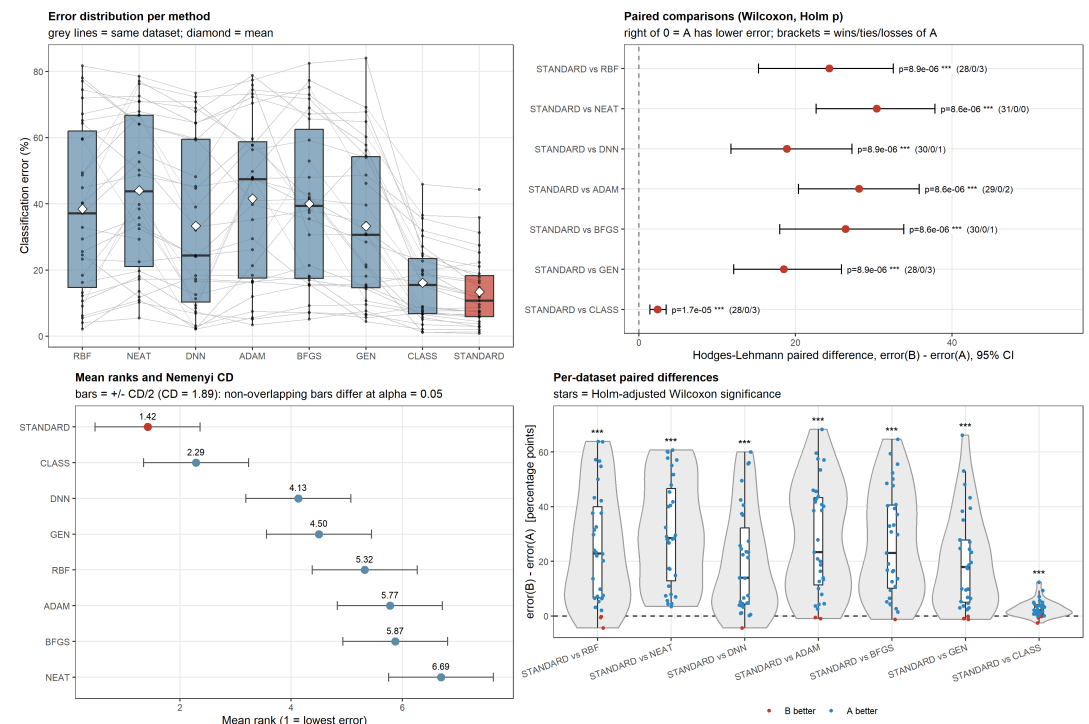
Table 2. Values for the parameters in experiments.

PARAMETER	MEANING	VALUE
N_c	Number of chromosomes	500
N_g	Number of allowed generations	500
p_s	Selection rate	0.9
p_m	Mutation rate	0.05
N_K	Crossover items	10
H	Number of weights for neural networks	10

Table 3. Comparison with other methods.

DATASET	RBF	NEAT	DNN	ADAM	BFGS	GEN	CLASS	STANDARD
ALCOHOL	49.38%	66.80%	39.04%	57.78%	41.50%	39.57%	16.70%	11.70%
APPENDICITIS	12.23%	17.20%	17.30%	16.50%	18.00%	18.10%	16.47%	12.80%
BEED	65.12%	67.94%	64.45%	70.38%	75.24%	46.19%	31.93%	27.45%
DERMATOLOGY	59.48%	32.43%	24.26%	26.14%	52.92%	30.58%	5.69%	2.80%
GENE	37.11%	52.68%	33.06%	35.17%	38.12%	34.38%	8.67%	7.29%
HAYES ROTH	64.36%	50.15%	44.65%	59.70%	37.33%	56.18%	24.31%	21.08%
HEPATITIS	18.38%	39.75%	16.65%	16.25%	21.38%	15.00%	15.50%	12.00%
HOUSEVOTES	6.13%	10.89%	3.13%	7.48%	7.13%	6.62%	3.47%	2.96%
IONOSPHERE	16.22%	19.67%	12.57%	16.64%	15.29%	15.14%	8.10%	8.74%
LYMOGRAPHY	25.50%	33.70%	24.07%	39.79%	35.43%	28.42%	22.67%	18.86%
NURSERY	77.10%	64.11%	59.73%	40.94%	41.40%	32.14%	19.65%	22.27%
OPTDIGITS	71.97%	72.56%	64.34%	72.32%	67.13%	67.81%	27.15%	14.77%
PENBASED	81.69%	78.58%	73.50%	77.41%	82.48%	84.01%	24.93%	17.79%
PHONEME	23.32%	22.43%	24.34%	29.43%	15.58%	15.55%	18.89%	16.82%
PIRVISION	24.58%	11.84%	18.28%	18.38%	16.94%	14.31%	6.53%	4.38%
SATIMAGE	40.19%	76.19%	72.30%	73.22%	71.73%	54.54%	18.47%	16.14%
SEGMENT	59.68%	66.72%	32.04%	49.75%	68.97%	57.72%	15.21%	9.57%
SHUTTLE	32.83%	29.32%	2.27%	21.29%	42.07%	20.86%	1.11%	1.28%
SPAMBASE	29.35%	35.77%	11.30%	48.05%	18.16%	6.37%	8.97%	7.28%
SPIRAL	44.87%	48.66%	45.24%	47.67%	47.99%	48.66%	35.65%	31.25%
SPLICE	40.27%	55.54%	48.18%	75.93%	30.64%	30.97%	7.68%	7.62%
STUDENT	5.49%	10.20%	6.93%	5.13%	7.14%	5.61%	6.86%	5.72%
TEXTURE	74.52%	70.82%	70.78%	56.33%	59.25%	54.00%	20.01%	10.71%
THYROID	10.62%	18.06%	7.42%	3.43%	5.18%	7.65%	1.24%	0.83%
TWONORM	2.18%	13.54%	2.16%	5.58%	9.22%	9.59%	6.77%	6.57%
VEHICLE	78.06%	64.05%	59.22%	74.41%	72.99%	60.24%	36.55%	35.83%
YEAST	67.14%	71.09%	68.87%	57.69%	77.28%	69.13%	45.95%	44.35%
Z_F_S	13.16%	38.41%	9.27%	47.81%	39.37%	10.73%	7.39%	6.00%
Z_O_N_F_S	48.70%	77.08%	67.80%	78.79%	65.67%	64.81%	30.32%	25.32%
ZO_NF_S	9.02%	43.75%	8.50%	47.43%	43.04%	21.54%	5.03%	3.69%
ZONF_S	4.03%	5.44%	2.52%	11.99%	15.62%	4.36%	2.04%	2.00%
AVERAGE	38.47%	44.04%	33.37%	41.57%	40.01%	33.25%	16.13%	13.42%

Friedman $\chi^2(7) = 121.47$, $p = 3.8\text{e-}23$ | Iman-Davenport $F = 38.15$, $p = 3.3\text{e-}34$ | Kendall $W = 0.56$

**Figure 1.** Statistical comparison of the proposed method (STANDARD) against the other classifiers over 31 datasets.

Note: Box plots: box = interquartile range, thick line = median, whiskers = $1.5 \times \text{IQR}$, diamond = mean, dots = datasets, grey lines join the same dataset. Forest plots: Hodges-Lehmann median paired difference in error, $\text{error}(B) - \text{error}(A)$, with 95% CI; right of the dashed line, A has lower error; labels give the Holm-adjusted Wilcoxon p -value and the wins/ties/losses of A. Rank panels: mean Friedman rank ($1 = \text{best}$) $\pm \text{CD}/2$ (Nemenyi); non-overlapping bars differ at $\alpha = 0.05$. Bottom right: per-dataset paired differences (blue: A better, red: B better) or, for \$N_K\$, the trend with bootstrap 95% CI. Stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

3.3. Crossover methods

Table 4. Crossover methods.

DATASET	STANDARD	TARGETED	TARGETEDWORST
ALCOHOL	11.70%	17.65%	14.04%
APPENDICITIS	12.80%	16.40%	15.40%
BEED	27.45%	29.48%	31.93%
DERMATOLOGY	2.80%	6.12%	5.14%
GENE	7.29%	9.05%	8.20%
HAYES ROTH	21.08%	22.00%	25.38%
HEPATITIS	12.00%	16.25%	16.25%
HOUSEVOTES	2.96%	3.72%	3.04%
IONOSPHERE	8.74%	7.77%	7.77%
LYMOGRAPHY	18.86%	25.00%	21.86%
NURSERY	22.27%	21.44%	21.02%
OPTDIGITS	14.77%	19.59%	19.88%
PENBASED	17.79%	15.10%	23.85%
PHONEME	16.82%	17.86%	17.86%
PIRVISION	4.38%	4.47%	6.53%
SATIMAGE	16.14%	17.52%	18.14%
SEGMENT	9.57%	12.86%	11.26%
SHUTTLE	1.28%	1.35%	0.94%
SPAMBASE	7.28%	8.22%	8.10%
SPIRAL	31.25%	35.30%	34.83%
SPLICE	7.62%	8.12%	8.15%
STUDENT	5.72%	5.92%	6.45%
TEXTURE	10.71%	17.23%	16.59%
THYROID	0.83%	1.35%	1.53%
TWONORM	6.57%	3.90%	5.00%
VEHICLE	35.83%	39.64%	37.03%
YEAST	44.35%	45.22%	47.77%
Z_F_S	6.00%	6.40%	7.33%
Z_O_N_F_S	25.32%	28.12%	30.66%
ZO_NF_S	3.69%	3.08%	4.04%
ZONF_S	2.00%	1.28%	2.04%
AVERAGE	13.42%	15.08%	15.42%

Friedman $\chi^2(2) = 22.13$, $p = 1.6e-05$ | Iman-Davenport $F = 16.65$, $p = 1.8e-06$ | Kendall $W = 0.36$

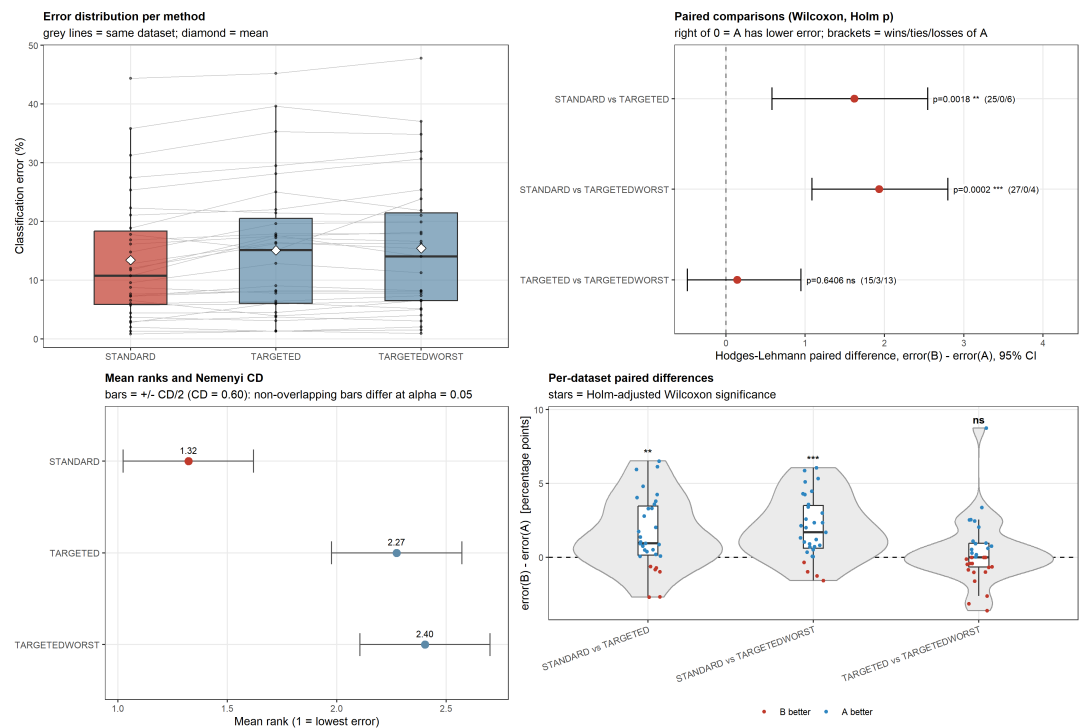


Figure 2. Statistical comparison of the three (STANDARD, TARGETED, TARGETEDWORST).

3.4. Number of crossing items

99

Table 5. Experiments with the number that participate in the crossover procedure.

DATASET	$N_K = 2$	$N_k = 5$	$N_K = 10$	$N_K = 15$
ALCOHOL	12.55%	13.19%	11.70%	15.11%
APPENDICITIS	15.60%	14.40%	12.80%	15.80%
BEED	27.30%	26.78%	27.45%	25.47%
DERMATOLOGY	4.74%	4.86%	2.80%	4.00%
GENE	8.39%	7.29%	7.29%	6.91%
HAYES ROTH	20.77%	19.54%	21.08%	21.39%
HEPATITIS	14.00%	11.50%	12.00%	10.00%
HOUSEVOTES	3.63%	2.78%	2.96%	3.28%
IONOSPHERE	7.60%	7.31%	8.74%	8.80%
LYMOGRAPHY	23.71%	23.00%	18.86%	20.57%
NURSERY	20.33%	22.16%	22.27%	18.76%
OPTDIGITS	17.16%	14.56%	14.77%	13.99%
PENBASED	19.95%	16.54%	17.79%	14.22%
PHONEME	16.99%	16.21%	16.82%	16.00%
PIRVISION	8.49%	6.33%	4.38%	5.35%
SATIMAGE	15.91%	15.04%	16.14%	14.93%
SEGMENT	11.95%	9.91%	9.57%	9.44%
SHUTTLE	1.15%	0.62%	1.28%	0.24%
SPAMBASE	7.70%	7.98%	7.28%	7.33%
SPIRAL	36.80%	33.55%	31.25%	30.35%
SPLICE	7.46%	7.18%	7.62%	7.34%
STUDENT	6.11%	4.92%	5.72%	6.70%
TEXTURE	12.87%	12.73%	10.71%	12.44%
THYROID	1.57%	1.06%	0.83%	0.76%
TWONORM	5.62%	5.31%	6.57%	8.01%
VEHICLE	37.62%	34.81%	35.83%	33.69%
YEAST	46.21%	45.69%	44.35%	44.01%
Z_F_S	7.00%	6.47%	6.00%	5.67%
Z_O_N_F_S	27.28%	27.00%	25.32%	23.80%
ZO_NF_S	3.68%	3.32%	3.69%	3.40%
ZONF_S	1.96%	1.44%	2.00%	1.16%
AVERAGE	14.58%	13.66%	13.42%	13.19%

Friedman $\chi^2(3) = 21.97$, $p = 6.6\text{e-}05$ | Iman-Davenport $F = 9.28$, $p = 2.1\text{e-}05$ | Kendall $W = 0.24$

Page trend (error falls as N_K rises): $z = 4.08$, one-sided $p = 2.3\text{e-}05$ | Spearman(N_K , error) < 0 in 23/29 datasets (sign-test $p = 0.0023$)

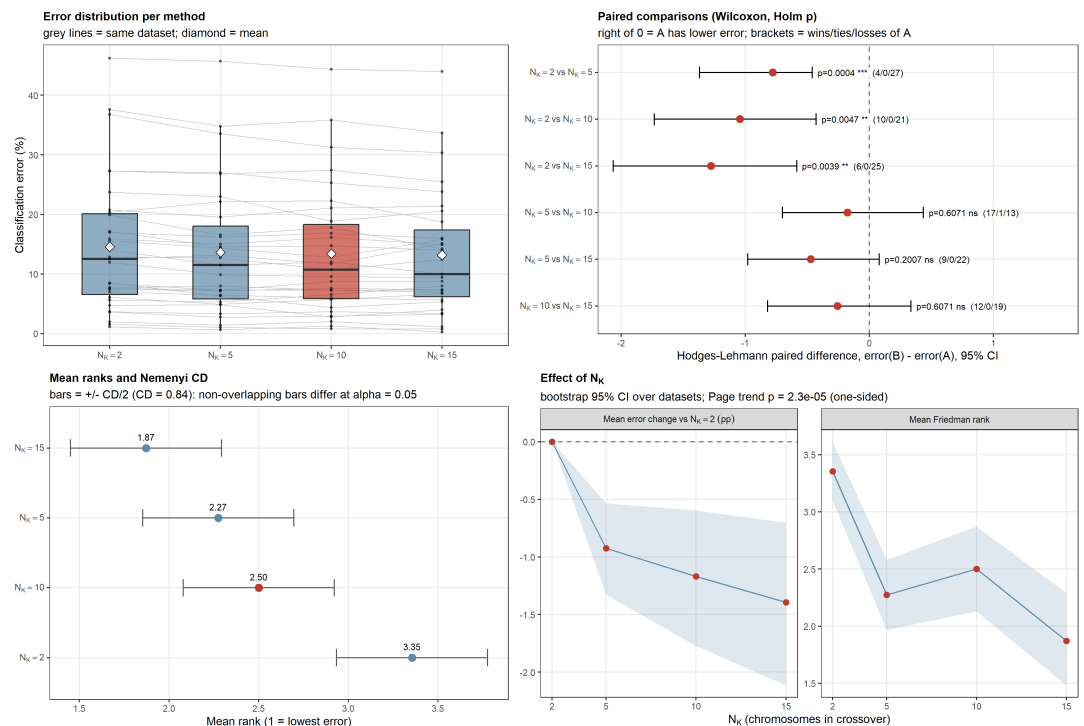


Figure 3. Effect of the number of chromosomes (N_K) participating in the crossover on the proposed method.

4. Conclusions

Author Contributions: G.G. and I.G.T. conducted the experiments, employing several datasets and provided the comparative experiments. V.C. a performed the statistical analysis and prepared the manuscript. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

References

1. M. Kelly, R. Longjohn, K. Nottingham, The UCI Machine Learning Repository, <https://archive.ics.uci.edu>.
2. J. Alcalá-Fdez, A. Fernandez, J. Luengo, J. Derrac, S. García, L. Sánchez, F. Herrera. KEEL Data-Mining Software Tool: Data Set Repository, Integration of Algorithms and Experimental Analysis Framework. *Journal of Multiple-Valued Logic and Soft Computing* 17, pp. 255-287, 2011.
3. Tzimourta, K.D.; Tsoulos, I.; Bilerio, I.T.; Tzallas, A.T.; Tsiouras, M.G.; Giannakeas, N. Direct Assessment of Alcohol Consumption in Mental State Using Brain Computer Interfaces and Grammatical Evolution. *Inventions* 2018, 3, 51.
4. Weiss, Sholom M. and Kulikowski, Casimir A., *Computer Systems That Learn: Classification and Prediction Methods from Statistics, Neural Nets, Machine Learning, and Expert Systems*, Morgan Kaufmann Publishers Inc, 1991.
5. G. Demiroz, H.A. Govenir, N. Ilter, Learning Differential Diagnosis of Erythematous-Squamous Diseases using Voting Feature Intervals, *Artificial Intelligence in Medicine*. **13**, pp. 147–165, 1998.

6. B. Hayes-Roth, B., F. Hayes-Roth. Concept learning and the recognition and classification of exemplars. *Journal of Verbal Learning and Verbal Behavior* **16**, pp. 321-338, 1977. 124
7. R.M. French, N. Chater, Using noise to compute error surfaces in connectionist networks: a novel means of reducing catastrophic forgetting, *Neural Comput.* **14**, pp. 1755-1769, 2002. 125
8. J.G. Dy , C.E. Brodley, Feature Selection for Unsupervised Learning, *The Journal of Machine Learning Research* **5**, pp 845–889, 2004. 126
9. S. J. Perantonis, V. Virvilis, Input Feature Extraction for Multilayered Perceptrons Using Supervised Principal Component Analysis, *Neural Processing Letters* **10**, pp 243–252, 1999. 127
10. G. Cestnik, I. Kononenko, I. Bratko, Assistant-86: A Knowledge-Elicitation Tool for Sophisticated Users. In: Bratko, I. and Lavrac, N., Eds., *Progress in Machine Learning*, Sigma Press, Wilmslow, pp. 31-45, 1987. 128
11. Olave, M., Rajkovic, V., & Bohanec, M. (1989). An application for admission in public school systems. *Expert Systems in Public Administration*, 1, 145-160. 129
12. Kaynak, C. (1995). Methods of combining multiple classifiers and their applications to handwritten digit recognition. Unpublished master's thesis, Bogazici University. 130
13. Emad-Ud-Din, M., & Wang, Y. (2023). Promoting occupancy detection accuracy using on-device lifelong learning. *IEEE Sensors Journal*, 23(9), 9595-9606. 131
14. M. Dash, H. Liu, P. Scheuermann, K. L. Tan, Fast hierarchical clustering and its validation, *Data & Knowledge Engineering* **44**, pp 109–138, 2003. 132
15. P. Cortez, A. M. Gonçalves Silva, Using data mining to predict secondary school student performance, In *Proceedings of 5th FUTURE BUSINESS TECHNOLOGY Conference (FUBUTEC 2008)* (pp. 5–12). EUROIS-ETI, 2008. 133
16. R. G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: dependence on recording region and brain state," *Physical Review E*, vol. 64, no. 6, Article ID 061907, 8 pages, 2001. 134
17. A. T. Tzallas, M. G. Tsipouras, and D. I. Fotiadis, "Automatic Seizure Detection Based on Time-Frequency Analysis and Artificial Neural Networks," *Computational Intelligence and Neuroscience*, vol. 2007, Article ID 80510, 13 pages, 2007. doi:10.1155/2007/80510 135